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Italy Trento San Pio X, 58 University of Trento

Riccardo Tedoldi

Computer Scientist & Enthusiast scholar and researcher

I am interested in the broad intersection between AI and healthcare to make an impact, improve people's lives and building a better tomorrow. As an enthusiastic researcher, I believe in the power of interdisciplinary teamwork. I enjoy engaging in meaningful discussions and exchanging ideas and perspectives with a positive and proactive mindset. The characteristics that truly encapsulate my essence are: ethical, exceptionally curious, creative, open-minded, a non-linear thinker, remarkably perseverant, and impeccably attentive to details. I have an immense passion for travel, hiking with my dog/running, capturing breathtaking photographs and listening music.

Personal data

Name: Riccardo **Surname:** Tedoldi **Date of birth:** 01/10/2000 **Citizenships:** Italian
Languages: Italian (mothertongue), English (fluent) **Occupation:** Student **M.Sc.:** Artificial Intelligent Systems
Personal web-site: r1cc4r2o.github.io/ **web-cv:** r1cc4r2o.github.io/cv/ **blog:** r1cc4r2o.github.io/blog/

Extra Curricula

Volunteer for the homeless — Trento, IT
The activity consists in distributing food and drinks to the homeless (Sant'Egidio Community) — *2023 - present*

Member of the UniTrento Polyphonic Choir — Trento, IT
The University Choir and Orchestra — *2023 - present*

Theatre school — Parma, IT
The activity entails gathering twice a week to immerse ourselves in the art of acting, actively preparing for an theatrical performance. Teatro al cerchio — *2016 - 2020*

Summer entertainer volunteer — Parma, IT
The activity consists in organizing games and activities for children and teenagers. St. Maria del Rosario — *2016 - 2019*

Member of the Choir Corale Verdi — Parma, IT
I was chosen from my elementary school choir to join the prestigious Verdi lyric choir coral. This incredible opportunity allowed me to participate in significant events held at renowned theaters, leaving a lasting impression on my musical journey. — *2008 - 2014*

Skills Summary

Legend: I have the basic of the underlined programming languages, while an exceptional proficiency in those **bolded**.

Languages: **Python**, C/**C++**, Bash, **LaTeX**, **Unix Terminal**, Prolog, CUDA, Java, R, Javascript, SQL

Tools: **PyTorch**, **PyTorch-Lightning**, PyTorch3D, TensorFlow, **Keras**, Scikit-learn, **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, MNE, OpenCV, **Git**, Docker, **Linux**, **Windows**, **MacOS**, Jupyter, Colab, Office, Adobe Illustrator, Adobe Lightroom, **Google Workspace**

Work experience

Research Intern — Parma, PR
Bioinformatics & Molecular Biology Intern — *Oct 2021 - Jun 2022*

- Clustering Algorithm
  - Developed a clustering algorithm that aimed to identify relevant atoms important for the protein binding.
  - The algorithm aimed to reduce the computational complexity of the previous one from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \cdot log(n))$ with comparable/better performance.
  - Supervisior: Dr. Marco Malatesta
- Neo-functionalization detection
  - Developed Deep Learning Transformers-based architectures (i.e. BERT, ViT, CCT, MaXViT) to detect whether mutations change the protein function. We worked with different protein representations (i.e. sequence, structure, embedding and more).
  - The architectures achieved on the average $\sim 90\%$ accuracy in neo-functionalization detection on the k-fold testset splits.
  - Supervisiors: PhD. Carlo DeRito, Dr. Marco Malatesta, Prof. Eleonora Iotti

Research

DhW: Unveiling the Shared Representation Emerging from Human Brain Responses — Trento, IT
*Seminar Paper, **R. Tedoldi**; Supervisor: U. Hasson; Code: (github)* — *Jul 2022*

An open source perspective on AI and alignment with the EU AI Act — Macao, CHN
*IJCAI AISafety 2023, Author: D. Calanzone, A. Coppari, **R. Tedoldi**, G. Olivato, C. Casonato* — *Jun 2022*

A Transformer-based Machine Learning Approach to Detect Protein Neofunctionalization — Parma, IT
*B.Sc. Thesis, **R. Tedoldi**; Supervisor: E. Iotti; Co-supervisors: C. De Rito, A. DalPalù* — *Jun 2022*

Education

University of Trento — Trento, TN
M. Sc.; Artificial Intelligence Systems *(Current GPA: **3.92/4**)* — *Sep 2022 - Present*
Full courses list: r1cc4r2o.github.io/cv/

University of Parma — Parma, PR
B. Sc.; Computer Science *GPA: **3.82/4*** ***First graduated in class of*** 141 — *Oct 2019 - Jul 2022*
Full courses list: r1cc4r2o.github.io/cv/

High-school Maria Luigia — Parma, PR
Scietific high-school cerfificate; — *Jun 2019*

Honors and Awards

- **Grant on the tuition fees of** 300€: The grant is awarded to the top 11 students of the Department of Computer Science of the University of Parma. I was awarded the grant in both consecutive years of my Bachelor's degree (2020/2021, 2021/2022).
- **First B.Sc. Degree of the Class**, Computer Science Jul 2022

Research Projects

Bioinformatics

UHC: A study on homologous protein latent encoding. (github) Jun 2023

- Relying on 15 human sequences we collected the best matching sequences in other species to identify similarities in their function. Supervisor: Prof. Jacopo Staiano

A Transformer-based Machine Learning Approach to Detect Protein Neofunctionalization Jun 2022

- Use cutting-edge Deep Learning architecture to perform protein Neofunctionalization detection.
  Supervisiors: PhD. Carlo DeRito, Dr. Marco Malatesta, Prof. Eleonora Iotti

Neuroscience

Decoding the hidden Waves:
Unveiling the Shared Representation Emerging from the Human Brain (github) Jul 2023

- Developed a framework to evaluate ML model on the task of brain decoding. We stated that pattern and similarities in the individual response, also emerge in the feature space. Supervisor: Prof. Uri Hasson

Computer Vision

Visual Grounding: Visual Grounding on RefCOCOg (github) Jul 2023

- Designed 2 solutions that achieved +4% AP compared to the baseline solution for the Deep Learning Course.
  Supervisiors: PhD. Alessandro Conti, Prof. Elisa Ricci

ShadowMe: Low-power 2D Motion Emulation and 3D Renders Learning (github) Jun 2023

- Developed a low-power pipeline with a web interface to emulate human motion in real-time/mp4. It works on iPad for 25/30fps. Supervisiors: PhD. Giulia Martinelli, Prof. Nicola Conci

Text-to-3D: Multi-view stable Point·E (github) Feb 2023

- Developed a multi-view conditioning to guide the generation of the point-cloud. Using multiple-views we state better divergence/chamfer distance scores (on ModelNet40 and ShapeNet).
  Supervisiors: PhD. Zeno Sambugaro, Prof. Francesco De Natale

Optimization

D2BL: Dynamic Distribution-driven Backpropagation Learning (github) Jun 2023

- Developed a mechanism to recombine the hidden features of the network. In a simple MLP with 4 layers, we improved the accuracy (+12%) by substituting the last layer with our mechanism.
  Supervisiors: PhD. Andrea Ferigo, Prof. Giovanni Iacca

Strategic Planning

The GoldGym: Business Plan of a Gym Dec 2021

- The document summarizes the goal of the business, the organization, financial progression and strategies. Also, we developed a website providing graphical plots of the client's trends to support the stack-holder in deciding.
  Supervisor: Prof. Armando Sternieri